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Gastric Residual Volume Assessment in Critically Ill Children

The GASTRIC-PICU Randomized Clinical Trial

Lyvonne N. Tume, PhD; Paul R. Mouncey, MSc; Julia M. Broomhall, PhD; Paloma Ferrando-Vivas, BA; Marzena Orzol, MSc; Irene Chang, MSc; Carly Au, BSc; Katherine L. Brown, MD; Julie Camsooksai, BSc; Nigel J. Hall, PhD; Jahara Khatun, BSc; Lamprini Lampro, MSc; Emma Laing, BSc; Lynne Latten, BSc; Luise V. Marino, PhD; Julie Menzies, PhD; Millie Parke, BSc; Zia Sadique, PhD; Hannah Sedgwick, MSc; Luregn J. Schlapbach, PhD; Frederic V. Valla, PhD; Lewis Veale, PhD; Kerry Woolfall, PhD; David A. Harrison, PhD; Mark J. Peters, PhD; for the GASTRIC-PICU Investigators of the Paediatric Critical Care Society Study Group (PCCS-SG)

IMPORTANCE Routine assessment of gastric residual volume (GRV) to guide enteral feeding in critically ill children is widespread but not based on evidence. Perceived high gastric volumes often lead to withholding feeds, impairing nutritional delivery.

OBJECTIVE To evaluate the effect of not routinely assessing GRV compared with assessments at least every 6 hours in children undergoing mechanical ventilation on the duration of mechanical ventilation and survival and achievement of nutritional targets.

DESIGN, SETTING, AND PARTICIPANTS A pragmatic, multicenter, randomized, noninferiority trial in 23 pediatric intensive care units (PICUs) in the UK and 1 in Switzerland. A total of 4700 children aged 0 to 16 years who were receiving invasive ventilation and starting enteral feeds were recruited between June 29, 2023, and December 7, 2025, with 30-day follow-up completed on January 6, 2026.

INTERVENTIONS Children were randomized (1:1) to receive usual care (GRV assessment every 6 hours) or no routine GRV assessment to guide enteral feeding. In the no routine GRV assessment group, feed tolerance was assessed using only clinical signs. All other enteral feeding practices followed local protocols.

MAIN OUTCOMES AND MEASURES The clinical co-primary outcome (noninferiority) was a composite of survival and days free from mechanical ventilation at 30 days. The nutritional co-primary outcome (superiority) was the percentage of children meeting energy requirements by 72 hours.

RESULTS Of the 4700 children randomized (2352 to the no routine GRV assessment group and 2348 to the usual care group), 4460 were included in the intention-to-treat analysis (median [IQR] age, 8 [1-44] months; 1925 [42.6%] females). No routine GRV assessment was noninferior to regular 6-hourly assessments for survival and days free from mechanical ventilation at 30 days (median [IQR], 25 [21-27] days in both groups; adjusted odds ratio [OR], 0.95 [95% CI, 0.86-1.05]). Results of the per-protocol analysis were consistent with the intention-to-treat analysis (adjusted OR, 1.01 [95% CI, 0.90-1.13]). The mean percentage of energy requirements achieved by 72 hours was 80.3% in the no routine GRV assessment group and 76.8% in the usual care group (adjusted mean difference, 3.2 [95% CI, 1.3-5.2] percentage points; $P < .001$).

CONCLUSIONS AND RELEVANCE Among critically ill children being enterally fed, not assessing GRV routinely was noninferior to regular assessments every 6 hours and significantly increased nutritional achievement at 72 hours.

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Author Affiliations: Author affiliations are listed at the end of this article.

Group Information: The GASTRIC-PICU Investigators of the Paediatric Critical Care Society Study Group (PCCS-SG) appear in Supplement 3.

Corresponding Author: Lyvonne N. Tume, PhD, Edge Hill University Faculty of Health Social Care and Medicine, St Helens Road, Ormskirk L39 4QP, United Kingdom (Lyvonne.tume@edgehill.ac.uk; I.n.tume@ljmu.ac.uk).

Adequate nutrition is a priority in critical illness.¹ However, children with critical illness typically receive less than half of their predicted energy requirements.^{2,3} Withholding of feed, based on perceived risks of ventilator-associated pneumonia (VAP) and necrotizing enterocolitis (NEC),⁴ contributes to this nutritional deficit. Routine assessment of gastric residual volume (GRV), by measuring the volume of fluid obtained by aspirating the gastric tube, is performed in almost all UK pediatric intensive care units (PICUs)⁵ and variably globally to predict tolerance of enteral feeding. Almost always the response to a perceived high GRV is to withhold feeds for a period of time. However, aspirate volume poorly reflects true GRV⁶ and aspirate volume thresholds used to predict feed tolerance are both variable and poorly evidenced.^{4,5}

A randomized clinical trial (RCT) in critically ill adults showed that not routinely assessing GRV had no impact on VAP, but facilitated delivery of nutrition.⁷ Systematic reviews have supported these findings.⁸ In critically ill children, a small observational study questioned the value of GRV assessment,⁹ but there have been no RCTs. There is greater potential for benefit of not assessing GRV in children due to lower energy reserves and higher metabolic demands compared with adults.¹⁰ Fear of vomiting and aspiration has been identified as the main reason for routinely assessing GRV in critically ill children.¹¹ Following a feasibility study,⁴ the GASTRIC-PICU trial¹² was conducted to evaluate whether not routinely assessing GRV to guide enteral feeding was noninferior in terms of a composite outcome of survival and days free from mechanical ventilation at 30 days and whether it increases the percentage of children who achieve estimated energy requirements by 72 hours compared with usual care.

Methods

Trial Design and Oversight

The GASTRIC-PICU RCT was a pragmatic, unblinded, multicenter, parallel-group, noninferiority trial. The study protocol was approved by London-Bloomsbury research ethics committee on May 10, 2023 (reference No. 23/LO/0284), and was published prior to completion of the trial.¹² The trial was registered at the United Kingdom (UK) clinical study registry (ISRCTN79668198) and the Swiss National Clinical trial Portal (SNCTPO00006200). The Consolidated Standards of Reporting Trials (CONSORT) reporting guideline was followed.

The UK National Institute for Health and Care Research Health Technology Assessment program funded the trial; independent data monitoring and trial steering committees were appointed. A planned interim analysis was conducted at 50% of the target sample size. The trial was sponsored and managed by the Intensive Care National Audit and Research Centre clinical trials unit.

Sites and Participants

The trial was conducted at 23 sites, comprising 23 PICUs in the UK National Health Service (with 1 site including 2 PICUs) and 1 in Switzerland. Children aged 0 (>37 weeks' gestational age)

Key Points

Question Is not routinely assessing gastric residual volume noninferior to regular assessment to guide enteral feeding in children with critical illness?

Findings In this randomized trial of 4700 children receiving invasive ventilation, not assessing gastric residual volume regularly was noninferior to usual care in terms of a composite of length of ventilation and mortality, but significantly increased the proportion of children reaching nutritional targets at 72 hours.

Meaning In children receiving invasive ventilation, the routine assessment of gastric residual volume did not alter clinical outcomes and impaired enteral feeding.

up to 16 years admitted to participating PICUs were included if they were enterally fed via the gastric route and were receiving invasive ventilation (Figure 1). Exclusion criteria were an end-of-life plan in place or death was perceived as imminent, receiving long-term ventilation via a tracheostomy, gut pathology contraindicating enteral feeding, or not being fed via the gastric route (eTable 1 in Supplement 2).

Randomization

Randomization was conducted using a concealed centralized web-based system based on a computer-generated randomization sequence and was stratified by site, age, and reason for admission. Children were randomized in a 1:1 ratio to undergo no routine GRV assessment or receive usual care within 24 hours of meeting all inclusion criteria. Because most UK PICUs initiated enteral nutrition early in the PICU admission,⁴ a research without prior consent approach was approved.^{13,14} Parents or legal guardians were approached for consent as soon as possible after randomization. If the parent or guardian refused consent, approvals allowed safety and primary outcome data to be collected along with data collected up to that point being retained unless the parent or guardian requested removal of all data. The full consent process is outlined in Supplement 1.

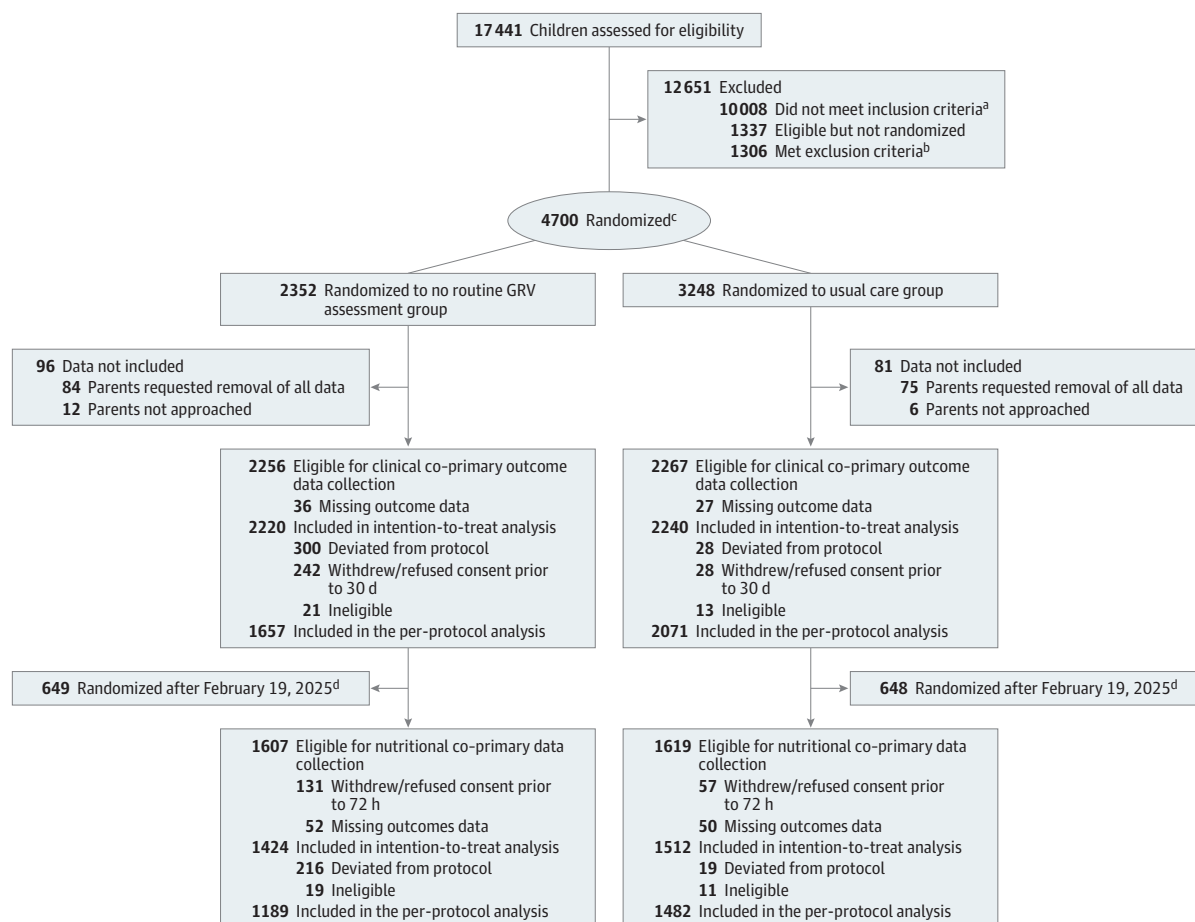
Trial Interventions

Following randomization, treatment started immediately. For participants in the no routine GRV assessment group, staff were required to not routinely assess GRV, using only clinical signs to evaluate feeding intolerance (eFigure 1 in Supplement 2). Staff were allowed to assess GRV in justifiable clinical situations, such as prior to a procedure, in instances of vomiting, or in response to clinical deterioration. For participants in the usual care group, staff were required to follow their standard local PICU practice, which included routine GRV assessments taken at least every 6 hours to guide enteral feeding. All other feeding practices followed local protocols and practices.

Study Outcomes

The co-primary outcomes were a composite of survival and days free from mechanical ventilation at 30 days (noninferiority) and the percentage of the child's estimated energy requirements achieved by 72 hours (superiority). Secondary outcomes were

Figure 1. Flow of Participants in the GASTRIC-PICU Randomized Clinical Trial



^aDid not meet inclusion criteria: aged <37 weeks' corrected gestational age or aged ≥ 16 years ($n = 443$); extubation planned in next 48 hours ($n = 6857$); no intention to start feeding via the gastric route ($n = 6182$). Participants may not meet >1 inclusion criteria.

^bMet exclusion criteria: postpyloric feeding or jejunostomy ($n = 171$); end-of-life care plan in place ($n = 127$); receiving long-term invasive mechanical ventilation ($n = 262$); current or recent gut pathology or surgery ($n = 452$); known to have

been enrolled in the GASTRIC-PICU trial in the last 6 months ($n = 346$). Participants may meet >1 exclusion criteria.

^cExcludes 90 participants randomized in error (duplicate randomizations/ previously randomized to the GASTRIC-PICU trial).

^dDate of blinded sample size reestimation where nutritional data collection was stopped.

time to achievement of target energy requirement, time to achievement of target protein requirement, diagnosis of VAP, diagnosis of NEC in infants (less than 12 months of age), duration of time with no enteral feed in the first 7 days, incidence of vomiting leading to feed stoppage in the first 7 days, documented health care-acquired infections, and length of PICU stay. The longer-term outcomes at 6 months and integrated economic evaluation will be reported separately. The detailed definitions of all outcomes can be found in [Supplement 1](#).

Sample Size Calculation

For the clinical co-primary outcome, a noninferiority margin of a lower limit for the odds ratio [OR] of 1/1.2 (0.833) was selected. This corresponded to a 0.8% absolute increase in mortality and a 12-hour difference in median duration of ventilation. For 90% power to detect noninferiority, based on the lower limit of a 2-sided 95% CI excluding this margin and using an out-

come distribution from the UK SANDWICH trial (4.2% mortality and log-normal distribution for duration of ventilation with a median of 2.9 days and lower quartile of 1.0 day),¹⁵ a sample size of 4000 was required. To retain power for a per-protocol analysis and allow for crossover of 10% and withdrawal of 5% of participants, we set a total target size of 4700. The nutritional co-primary outcome was anticipated to be approximately normally distributed, with SDs within each group between 20% and 40%.⁹ However, because of this uncertainty in the SD, a blinded sample size reestimation was undertaken at the interim analysis, according to a prespecified statistical analysis plan. The sample size in the intention-to-treat population required to provide 90% power at a 2-sided P value of less than .05 to detect a 4% absolute difference in energy intake based on the observed pooled SD was 2258. This effect size was chosen to detect a small (Cohen d of 0.2) to very small (Cohen d of 0.1) effect based on the anticipated range of the SD.

Table 1. Baseline Participant Characteristics

Characteristics	No routine GRV assessment (n = 2256)	Usual care (n = 2267)
Age, No./total No. (%), mo ^a		
<1	417/2256 (18.5)	422/2267 (18.6)
≥1 to <12	892/2256 (39.5)	899/2267 (39.7)
≥12	947/2256 (42.0)	946/2267 (41.7)
Age, mo ^b	n = 2253	n = 2261
Mean (SD)	34.1 (52.6)	34.0 (51.7)
Median (IQR)	8 (1-43)	8 (1-45)
Sex, No./total No. (%) ^b		
Female	950/2253 (42.2)	975/2261 (43.1)
Male	1303/2253 (57.8)	1286/2261 (56.9)
Ethnicity, No./total No. (%) ^{c,d}		
Asian	260/1877 (13.9)	296/1982 (14.9)
Black	113/1877 (6.0)	125/1982 (6.3)
White	1076/1877 (57.3)	1139/1982 (57.5)
Multiple ethnicities	71/1877 (3.8)	87/1982 (4.4)
Other	71/1877 (3.8)	57/1982 (2.9)
Not stated	286/1877 (15.2)	279/1982 (14.0)
Quintile of IMD, No./total No. (%) ^{c,e}		
1 (Least deprived)	210/1567 (13.4)	217/1652 (13.1)
2	254/1567 (16.2)	291/1652 (17.6)
3	307/1567 (19.6)	304/1652 (18.4)
4	374/1567 (23.9)	415/1652 (25.1)
5 (Most deprived)	422/1567 (26.9)	425/1652 (25.7)
Weight, kg ^b	n = 2253	n = 2261
Mean (SD)	13.8 (16.6)	13.5 (15.6)
Median (IQR)	8 (4-15)	8 (4-15)
Child feeds normally prior to randomization, No./total No. (%) ^b	1814/2239 (81.0)	1831/2252 (81.3)
Intended feeding route, No./total No. (%) ^b		
Gastric tube	2113/2247 (94.0)	2092/2254 (92.8)
Gastrostomy	134/2247 (6.0)	162/2254 (7.2)
Cardiac reason for admission, No./total No. (%) ^a	474/2256 (21.0)	489/2267 (21.6)
PIM-3 score at admission (%) ^{c,f}	n = 1880	n = 1983
Mean (SD)	4.6 (7.9)	4.7 (8.9)
Median (IQR)	1 (1-5)	2 (1-5)
Planned admission, No./total No. (%) ^c	373/1823 (20.5)	424/1891 (22.4)
Length of PICU stay prior to randomization, median (IQR), h ^b	18 (8-37)	18 (8-36)
Length of ventilation prior to randomization, median (IQR), h ^b	19 (10-34)	20 (10-35)

^a From randomization system.^b From the electronic case report form.^c From PICANet linkage.^d Child's ethnic origin as recorded in the medical records defined according to standard National Health Service ethnic categories and codes and Ethnic Category 2021 categories.^e Index of multiple deprivation (IMD) is an index of relative deprivation based on postcode. IMD is reported by quintiles, with a lower quintile indicating less deprivation.^f Paediatric Index of Mortality 3 (PIM-3) score is an index of mortality predicting the risk of mortality at admission to pediatric intensive care unit (PICU). The score ranges from 0% to 100%, with a lower score indicating a lower probability of mortality.

Statistical Analysis

The clinical co-primary outcome was analyzed using a proportional odds logistic regression model, with death during the first 30 days following randomization ranked as the worst outcome and surviving participants assigned values according to their total calendar days free from mechanical ventilation during the first 30 calendar days following randomization. The model was adjusted for the stratification variables, which were selected a priori based on their established relationship with outcome for children with critical illness. Multiple imputation was not required for any of the baseline variables due to the high completeness. The primary effect estimate is the adjusted OR, which was calculated in both the intention-to-treat and per-protocol populations with 2-sided 95% CIs. Non-inferiority would be declared if the lower limit of the CI was not less than 1/1.2 in both the intention-to-treat and per-protocol populations. Secondary analyses tested the same outcome for superiority and inferiority. The 2 separate components of the composite clinical co-primary outcome are reported by group. The nutritional co-primary outcome was analyzed using a linear regression model adjusted for the same baseline variables as the clinical co-primary outcome. The primary effect estimate is the difference in mean percentage of target achieved between the groups in the intention-to-treat population, which was tested for superiority and reported with a 2-sided 95% CI. Both co-primary outcomes are reported in 2 clinically relevant subgroups: reason for admission (cardiac, respiratory, other) and Paediatric Index of Mortality 3 score at admission.

All secondary outcomes are reported in both the intention-to-treat and per-protocol populations unless otherwise specified. Time-to-event outcomes were analyzed using Cox proportional hazards regression. Participants who did not achieve nutritional targets were censored at death, discharge from PICU, or day 7 following randomization (whichever occurred first). Binary end points were compared using adjusted logistic regression. The diagnosis of VAP is reported as a rate per 1000 hours of ventilation and was analyzed using Poisson regression. Continuous end points were analyzed using adjusted linear regression. Length of stay end points were analyzed using Wilcoxon rank sum tests, stratified by survival. Ethnic origin was collected to ensure inclusivity of the research, using documentation in the child's medical records defined according to standard UK National Health Service categories. Statistical analyses were conducted in Stata/MP version 18.0 (StataCorp).

Results

Sites and Participants

Between June 29, 2023, and December 7, 2025, a total of 17 441 critically ill children receiving invasive mechanical ventilation were screened at 23 sites, of whom 5982 were potentially eligible for inclusion and 4700 were randomized (Figure 1, Table 1, and eFigure 2 in Supplement 2). There were 159 participants (3.4%) whose parents requested removal of all data, 18 participants (0.4%) whose parents were not approached for

Table 2. Primary and Secondary Outcomes^a

Outcome	No routine GRV assessment	Usual care	Risk difference (95% CI)	Effect estimate (95% CI)	P value
Co-primary outcomes					
Survival and days free of mechanical ventilation, median (IQR) [No.], d					
Intention-to-treat population	25 (21-27) [2220]	25 (21-27) [2240]		OR, 0.95 (0.86-1.05)	.006 ^b
Per-protocol population	26 (22-27) [1657]	26 (22-27) [2071]		OR, 1.01 (0.90-1.13)	<.001 ^b
Complier average causal effect				OR, 0.99 (0.89-1.10)	<.001 ^b
Participants with nutrition achieved by 72 h, mean (SD) [No.], %					
Intention-to-treat population	80.3 (28.2) [1424]	76.8 (30.5) [1512]		MD, 3.24 (1.29-5.19)	<.001
Per-protocol population	81.1 (27.7) [1189]	77.0 (30.5) [1482]		MD, 3.34 (1.30-5.39)	<.001
Complier average causal effect				MD, 3.90 (1.55-6.25)	<.001
Secondary outcomes at day 8 (intention-to-treat population)					
Incidence of vomiting leading to feed stoppage, No./total No. (%)	204/1992 (10.2)	190/2105 (9.0)	1.2 (−0.5 to 3.0)	OR, 1.16 (0.94-1.43)	.17
Time to achievement of target calories, median (IQR) [No.], d	6.0 (5.0 to 6.0) [1417]	6.0 (5.0 to 6.0) [1518]	1.2 (−0.5 to 3.0) ^c	HR, 1.01 (0.91-1.12)	.86
Time to achievement of target protein, median (IQR) [No.], d	0.3 (−3.0 to 3.6) [1416] ^d	0.9 (−3.0 to 4.7) [1518] ^c	1.2 (−0.5 to 3.0) ^c	HR, 1.03 (0.90-1.17)	.68
Time with no enteral feed, mean (SD) [No.], h	20.7 (21.7) [1992]	22.7 (23.5) [2105]		MD, −1.7 (−3.1 to −0.4)	<.001
Secondary outcomes at PICU discharge (capped at 30 d; intention-to-treat population)					
Diagnosis of necrotizing enterocolitis, No./total No. (%)	30/1158 (2.6)	31/1228 (2.5)	0.1 (−1.1 to 1.3)	OR, 1.05 (0.63-1.76)	.85
Diagnosis of VAP, No./total No. (%)	120/2007 (6.0)	114/2128 (5.4)	0.6 (−0.8 to 1.9)	OR, 1.12 (0.85-1.47)	.42
VAP rate per 1000 h of ventilation, No./total No. (%)	0.37	0.34		IRR, 1.09 (0.84-1.41)	.52
Other documented health care-acquired infections, No./total No. (%)	222/2007 (11.0)	252/2128 (11.8)	−0.7 (−2.5 to 1.2)	OR, 0.93 (0.76-1.13)	.47
Secondary outcomes at discharge (intention-to-treat population)					
Length of PICU stay, median (IQR) [No.], d					
Survivors	5 (3-11) [1922]	5 (3-10) [2038]		PI, 0.50 (0.48-0.52)	.79
Nonsurvivors	14 (5-36) [86]	11 (4-24) [87]		PI, 0.42 (0.34-0.51)	.08
Length of hospital stay, median (IQR) [No.], d					
Survivors	13 (7-28) [1853]	13 (7-26) [1953]		PI, 0.50 (0.48-0.52) ^d	.86
Nonsurvivors	17 (6-63) [99]	15 (5-37) [109]		PI, 0.46 (0.38-0.54) ^d	.35

Abbreviations: GRV, gastric residual volume; HR, hazards ratio; IRR, incidence rate ratio; MD, mean difference; OR, odds ratio; PI, probabilistic index from Wilcoxon rank sum test; VAP, ventilator-acquired pneumonia.

^a Analyses are adjusted for stratification variables, except for analyses of length of pediatric intensive care unit (PICU) and hospital stay, which are unadjusted.

^b Based on noninferiority margin of 1/1.2.

^c Risk difference at 8 days post randomization.

^d Less than 50% of participants achieved target protein.

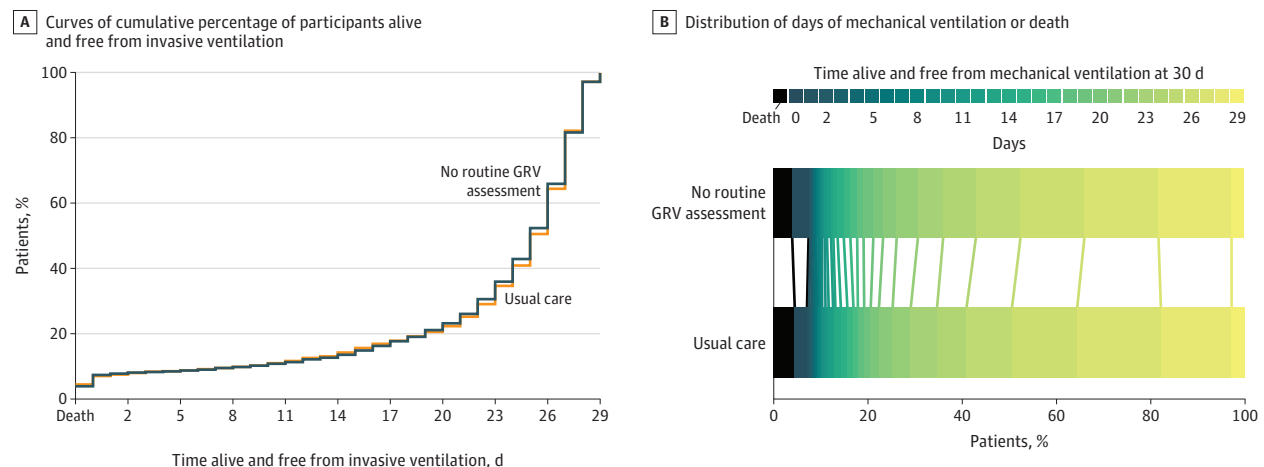
consent, and 63 participants (1.3%) with missing outcome data, leaving 4460 (2220 in the no routine GRV assessment group and 2240 in the usual care group) participants included in the clinical co-primary outcome analysis (eTable 2 in [Supplement 2](#)). Data collection for the nutritional co-primary outcome analysis stopped on February 19, 2025, following the pre-specified blinded sample size reestimation, which indicated sufficient sample size. An additional 188 parents (4.0%) refused or withdrew consent or opted out of their child's data collection, leaving 2936 (1424 in the no routine GRV assessment group and 1512 in the usual care group) in the analysis of the nutritional co-primary outcome. Participants whose parents had refused or withdrawn consent were also excluded from analyses of secondary outcomes. Thirty-day follow-up was completed on January 6, 2026.

Patient baseline characteristics were similar between the 2 groups (Table 1). More than half of participants (58.1%) were younger than 1 year, with a median age of 8 months in each group. The majority of participants were male (57.2%), were feeding or eating normally prior to randomization (80.6%), and were admitted for a noncardiac reason (78.7%). Participants were representative of the population admitted to PICUs in the UK.¹⁶ The majority of participants (58.1%) received continuous enteral feeding as their initial mode of feeding (eTable 3 in [Supplement 2](#)).

GRV Assessments

A total of 48 516 GRV assessments were performed in 2208 participants, equating to 3.8 assessments per 24 hours, in the usual care group. In the no routine GRV assessment group, 5903 GRV

Figure 2. Line and Bar Graphs of Distribution of the Clinical Co-Primary Outcome of Survival and Days Free of Mechanical Ventilation at 30 Days



A, Curves that increase more steeply indicate a less favorable distribution in the number of days of mechanical ventilation. B, Blue represents worse outcomes and yellow represents better.

assessments were performed in 1267 participants, equating to 0.5 assessments per 24 hours (eTable 4 in Supplement 2). Adherence was monitored in the first 4 charting days following randomization. In-line with the protocol, 363 participants (16.1%) in the no routine GRV assessment group had 1 or more GRV assessments for specified clinical reasons. In addition, 357 participants (15.9%) in the no routine GRV assessment group had 1 or more GRV assessments for reasons not specified in the protocol or, most commonly, for an unknown or undocumented reason. In the usual care group, 29 participants (1.3%) had 1 or more calendar days with no GRV assessment (eTable 5 in Supplement 2). In the no routine GRV assessment group, 80 participants (4.0%) had their feed stopped due to the GRV assessment compared with 406 (19.3%) in the usual care group (eTable 6 in Supplement 2).

Primary and Secondary Outcomes

The median (IQR) survival and days free of mechanical ventilation at 30 days was 25 (21-27) days in both groups (adjusted OR, 0.95 [95% CI, 0.86-1.05]; Table 2 and Figure 2 and eTables 7 and 8 in Supplement 2). The lower bound of the 95% CI fell within the prespecified noninferiority margin of 1/1.2 ($P < .001$), confirming the noninferiority of the intervention. Similar results were seen in the per-protocol analysis (adjusted OR, 1.01 [95% CI, 0.90-1.13]; $P = .001$). The mean percentage of estimated energy requirements by 72 hours after randomization was 80.3% in the no routine GRV assessment group and 76.8% in the usual care group (adjusted mean difference, 3.2 [95% CI, 1.3-5.2] percentage points; $P < .001$).

The duration of time with no enteral feed in the first 7 days was significantly lower in the no routine GRV assessment group (adjusted mean difference, -1.7 [95% CI, -3.1 to -0.4] hours; $P < .001$). There were no significant differences between the groups for any of the other secondary outcomes, including incidence of vomiting leading to feed stoppage in the first 7 days (adjusted OR, 1.16 [95% CI, 0.94-1.43]; $P = .17$), time to achieve-

ment of target calories (adjusted HR, 1.01 [95% CI, 0.91-1.12]; $P = .86$), and time to achievement of target protein (adjusted HR, 1.03 [95% CI, 0.90-1.17]; $P = .68$). The diagnosis of NEC was not significantly different between the groups (2.6% vs 2.5%; adjusted OR, 1.05 [95% CI, 0.63-1.76]; $P = .85$). Diagnosis of VAP was similar in the no routine GRV assessment and usual care groups (6.0% vs 5.4%; adjusted OR, 1.12 [95% CI, 0.85-1.47]; $P = .42$). Secondary outcomes were consistent in the per-protocol population (eTable 9 in Supplement 2). Three serious adverse events were reported: 2 (ischemic colitis; transverse colon perforation) in the no routine GRV assessment group and 1 (cecal perforation) in the usual care group.

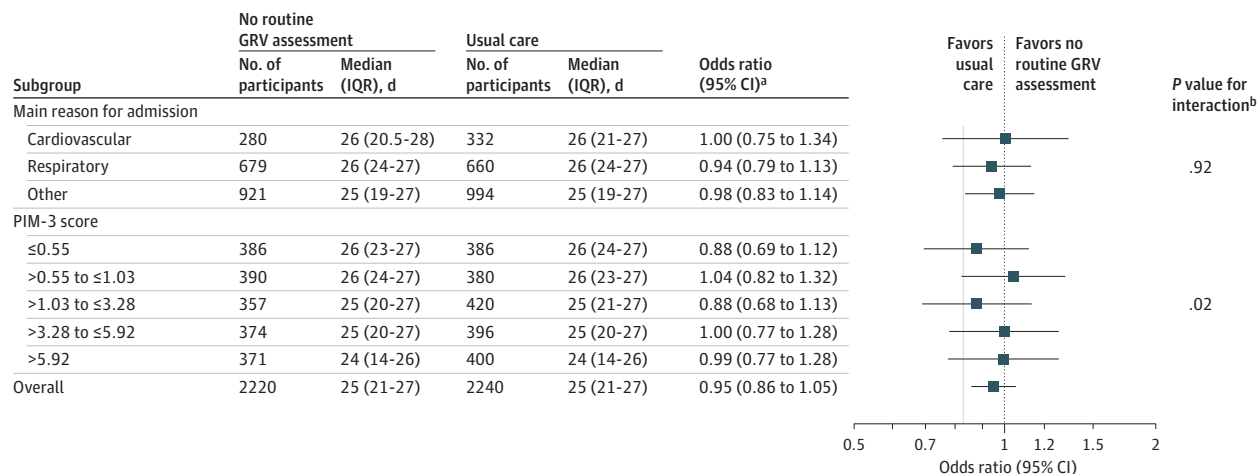
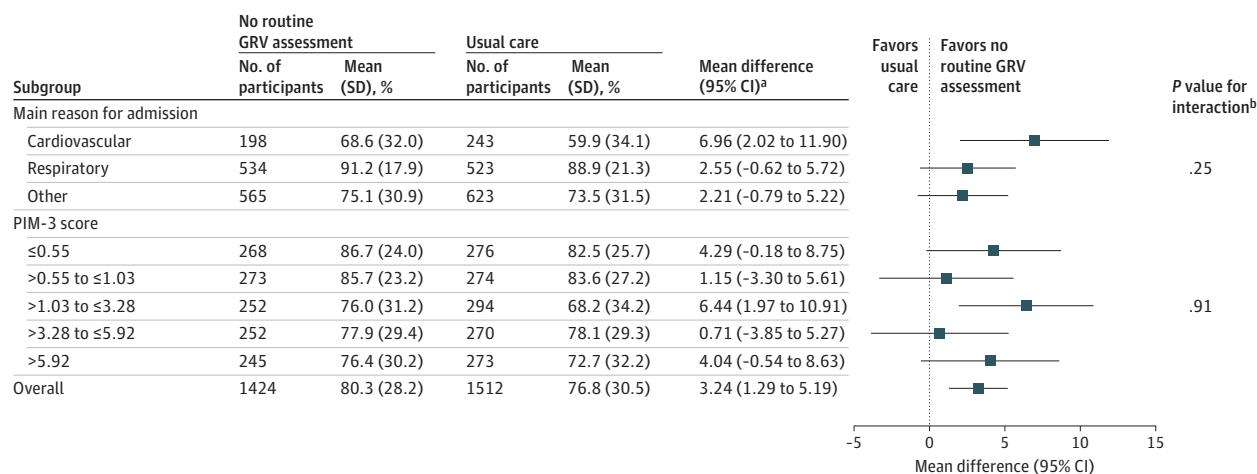
There was no evidence of heterogeneity by reason for admission and Paediatric Index of Mortality 3 score at admission for either co-primary outcomes (Figure 3).

Discussion

Among children receiving invasive ventilation admitted to UK PICUs, not routinely assessing GRV did not worsen the composite outcome of survival and days free of mechanical ventilation compared with usual care. In addition, omitting this practice significantly increased the percentage of the child's estimated energy requirements achieved by 72 hours. Although this effect was small, any improvements to reduce cumulative energy deficits are believed to be beneficial.

There was no clinically significant harm associated with not routinely assessing GRV detected. This included areas of prior concern including diagnosis of VAP and NEC. These findings support widespread deimplementation of the practice of routine GRV assessment in children with critical illness. However, there remain justifiable reasons to assess GRV based on clinical assessment. These results are consistent with a previous multicenter trial in adults with critical illness,⁷ single-center trials in preterm neonates,^{17,18} and observational data

Figure 3. Dot and Whisker Plot of Subgroup Analyses of the Co-Primary Outcomes

A Survival and days free from mechanical ventilation**B** Percentage of participants with energy targets achieved by 72 hours

A, The line at odds ratio of 1/1.2 indicates the noninferiority margin used in the trial. Where the lower limit of the 95% CI does not cross the noninferiority margin line, no routine GRV assessment is deemed noninferior. Analyses were adjusted for main reason for admission to PICU, age group at randomization and site as a random effect. Continuous interactions are illustrated using

quintiles of the continuous variable. GRV indicates gastric residual volume; PIM-3, Paediatric Index of Mortality 3.

^aOdds ratio from proportional odds logistic regression.

^bMean difference from linear regression model.

in children.⁹ A multicenter noninferiority trial in 452 adults showed no increase in VAP when GRV was not assessed alongside significantly greater nutritional attainment.⁷ Two single-center trials involving 258 preterm neonates showed no harm when GRV was not assessed, but did not detect a change in the time to achieve full feeds,¹⁷ although weight gain and overall nutritional delivery were improved.¹⁸ A large, international trial (neoGASTRIC)¹⁹ is currently assessing the risks and benefits of routine GRV assessment in 7040 preterm neonates; recruitment is completed and results are expected in late 2026.

The strengths of the current trial include both the total number of participants recruited (the largest individually randomized trial undertaken in PICU) and the representativeness of this population to the UK population of children with critical illness. The trial was performed in 23 of the 27

UK PICUs and 1 in Switzerland, and a high proportion of eligible patients were randomized. As with other recent trials, the research without prior consent model facilitated this inclusiveness.^{20,21} The end points were selected with extensive consultation with professionals and patients and families as being of direct relevance to their experience of the PICU. The duration of ventilation and 30-day mortality in the usual care group were similar to the pretrial estimates. The sample size was sufficient to exclude a 0.8% absolute change in mortality and a 12-hour difference in median duration of ventilation. The sample size required to determine noninferiority of this outcome meant there was sufficient power to capture the impact on delivery of nutrition. Adherence was high, leading to a high separation between the groups in terms of GRV assessments.

Limitations

This study has limitations. First is the presence of postrandomization exclusions. This has been seen in previous pragmatic trials and is likely linked to the use of the research without prior consent approach. Although the total numbers were small compared with the trial population, it should be noted that consent was declined more frequently in the intervention than control group. As a mitigation for this, ethical approvals allowed the inclusion of primary outcome and safety data for nonconsented children. Second, as an unblinded study, there was potential for bias from the bedside nurses' threshold for nonroutine aspirations of gastric tubes varying. Although the adherence data suggest that this was not common, the possibility that staff chose to aspirate to confirm the nasogastric tube position at different thresholds in the no routine GRV assessment group cannot be excluded. Third, only

children receiving invasive ventilation were included, and future randomized trials should examine whether GRV assessment is helpful for those receiving noninvasive respiratory support. These results, alongside similar findings in adult trials, might arise either from the true GRV not predicting risk and benefits of enteral feeding effectively, or from the volume that can be aspirated via a gastric tube being an insufficiently precise measure of that true volume.

Conclusions

Among critically ill children receiving mechanical ventilation in the PICU, routine GRV assessments used to guide enteral feeding do not improve outcomes and reduce the achievement of nutritional goals.

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Author Affiliations: Faculty of Health, Social Care & Medicine, Edge Hill University, Ormskirk, United Kingdom (Tume); Faculty of Health, Innovation, Technology and Science, Liverpool John Moores University, Liverpool, United Kingdom (Tume); Clinical Trials Unit, Intensive Care National Audit & Research Centre, London, United Kingdom (Mouncey, Broomhall, Ferrando-Vivas, Orzol, Chang, Au, Camsooksai, Khatun, Lampro, Laing, Parke, Sedgwick, Harrison); Great Ormond St Hospital NHS Foundation Trust and NIHR Biomedical Research Centre, London, United Kingdom (Brown); Faculty of Medicine, University of Southampton, Southampton, United Kingdom (Hall); Alder Hey Children's Foundation NHS Trust, Liverpool, United Kingdom (Latten, Marino); Paediatric Intensive Care, University Hospitals Bristol and Weston NHS Foundation Trust, Bristol, United Kingdom (Menzies); College of Health, Science and Society, University of the West of England, Bristol, United Kingdom (Menzies); Faculty of Public Health and Policy, London School of Hygiene and Tropical Medicine, London, United Kingdom (Sadique); Department of Intensive Care and Neonatology, University Children's Hospital Zurich, Zurich, Switzerland (Schlapbach); Child Health Research Centre, The University of Queensland, Brisbane, Australia (Schlapbach); Nutrition Support Team Hospices Civils de Lyon, Lyon, France (Valla); Parent Representative, United Kingdom (Veale); Institute of Population Health, The University of Liverpool, Liverpool, United Kingdom (Woolfall); University College London, Great Ormond St Institute of Child Health, London, United Kingdom (Peters); Paediatric Intensive Care Unit, Great Ormond St Hospital NHS Foundation Trust, London, United Kingdom (Peters).

Author Contributions: Drs Broomhall, Ferrando-Vivas, and Harrison had full access to all of the data in the study and take responsibility for the integrity of the data and the accuracy of the data analysis.

Concept and design: Tume, Mouncey, Orzol, Brown, Camsooksai, Hall, Laing, Latten, Marino, Menzies, Parke, Sadique, Sedgwick, Schlapbach, Valla, Veale, Woolfall, Harrison, Peters.

Acquisition, analysis, or interpretation of data:

Tume, Mouncey, Broomhall, Ferrando-Vivas, Orzol, Chang, Au, Brown, Camsooksai, Khatun, Lampro, Laing, Parke, Sadique, Sedgwick, Schlapbach, Valla, Harrison, Peters.

Drafting of the manuscript: Tume, Mouncey, Broomhall, Ferrando-Vivas, Orzol, Camsooksai, Parke, Sedgwick, Peters.

Critical review of the manuscript for important intellectual content: Tume, Broomhall, Orzol, Chang, Au, Brown, Hall, Khatun, Lampro, Laing, Latten, Marino, Menzies, Sadique, Schlapbach, Valla, Veale, Woolfall, Harrison, Peters.

Statistical analysis: Broomhall, Ferrando-Vivas.

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Administrative, technical, or material support:

Tume, Orzol, Chang, Au, Brown, Camsooksai, Khatun, Lampro, Laing, Latten, Marino, Menzies, Parke, Sadique, Sedgwick, Schlapbach, Valla, Peters.

Supervision: Tume, Mouncey, Orzol, Brown, Hall, Latten, Valla, Harrison, Peters.

Other - trial processes: Woolfall.

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Additional Contributions: Data monitoring and ethics committee (independent): Kerry Hood, PhD (chair); Bronagh Blackwood, PhD; and Kevin Morris, PhD. Trial steering committee (independent unless stated): Chris Gale (chair; independent), PhD; Emma Gentles, MSc; Lisa McIlMurray, PhD; Nilesh Mehta, PhD; Catherine Hewitt, PhD; Miriam Musey; Lyvonne Tume, PhD (nonindependent); Mark Peters (nonindependent), PhD; and Nicola Kenworthy.

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